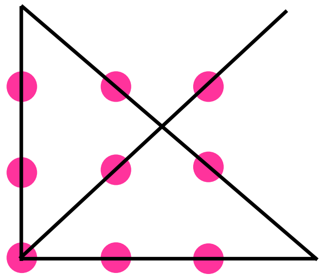
Page 72–73: Puzzling shapes

1. Most people think the answer is 32, since it seems to double each time. In fact, it's 31.

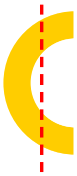
2. All the letters above the division are made of straight lines, while all the letters below contain curves.

3.



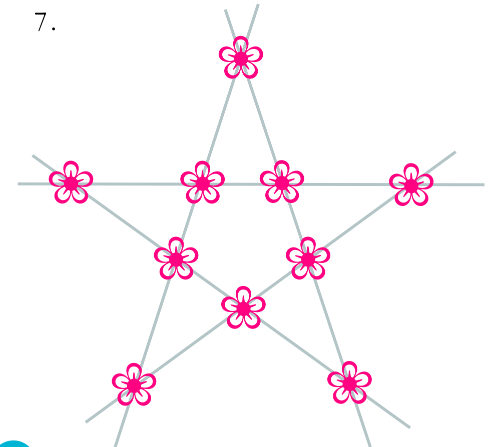
4. By making two vertical cuts at right angles and one horizontal cut right through the cake.

5. First make a horizontal cut, as though you're slicing open a bagel. Then make a vertical cut to slice the circle into two semicircles. Finally, stack one on top of the other and make a cut like this:

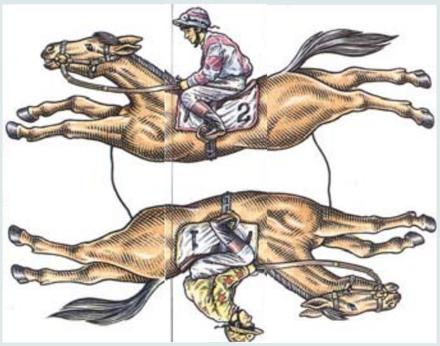


6. Poke your finger through the handle and give it a push!!!

7.

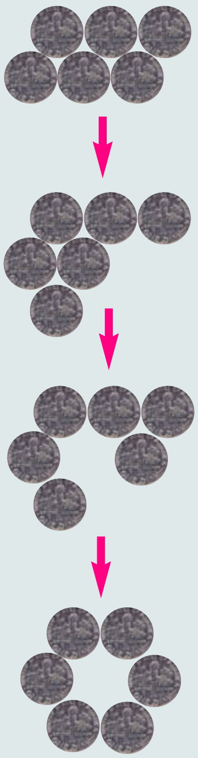


8.



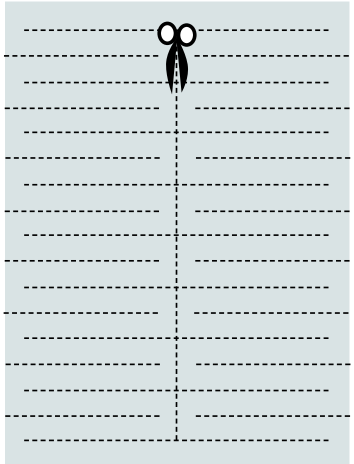
9. The ant can walk around the octahedron but not the cube or the tetrahedron. The journey is impossible if more than two corners of a shape have an *odd number* of connections to other corners. For a similar puzzle, see the Königsberg bridges, page 71.

10. There are 24 ways of solving the sliding coin puzzle.
Here's one:



11. Green.

12.



Page 82–83: Logic

Chessboard

It's impossible to cover all the squares with dominoes. Each domino must lie on both a black and a white square, so the dominoes will cover an equal number of each. But since the missing squares are both black, there will be two spare white squares that can't be covered.

Who shaves the barber?

Nobody – she doesn't shave!

The prisoner

The prisoner can't know which day the hanging will take place on. Therefore, he can't be hung on the last day, because he'd know the day before. Likewise, he can't be hung the day before last, because he'd know the day before that. Working backwards, the same thing applies to every day, so the prisoner can't be hung at all!

The tiger

If the woman says "you will let the child go", the tiger can do